

Concept 5 | Applications of Integration to Geometry

English Student Edition • Focused formulas and guided practice

Formula opener

Area

$$A = \int_a^b (y_{top} - y_{bottom}) dx$$

Volume

$$V = \pi \int_a^b y^2 dx$$

Bounds

$$a \leq x \leq b$$

Classified Practice

Applications of Integration to Geometry

Q001 Written

Find the area bounded by $y = 2x^2 + 1$, the x -axis, $x = -2$, and $x = 1$.

Q002 Written

Find the area bounded by $y = x^2 + x - 6$, the x -axis, $x = -2$, and $x = 1$.

Classified Practice

Applications of Integration to Geometry

Q003 Written

Find the area bounded by $y = -x^2 + 4$ and the x -axis.

Q004 Written**Find the area under $y = -x^2 - 2x + 3$ and above the x -axis.**

Classified Practice

Applications of Integration to Geometry

Q005 Written

Find the area between $y = x^2 - 2$ and $y = -2x + 1$.

Q006 Written**Find the area between $y = x^2 + 2x - 4$ and $y = -x^2 + 2x + 4$.**

Classified Practice

Applications of Integration to Geometry

Q007 Written

Find the area between $y = -x^2 + 4x$ and $y = x$.

Q008 Written**Find the area between $y = -x^2 + 8$ and $y = 2x$ for $-2 \leq x \leq 3$.**

Classified Practice

Applications of Integration to Geometry

Q009 Written

Find the area bounded by $y = x^3 - 5x^2 + 6x$ and the x -axis.

Q010 Written

Find the area bounded by $y = -x^3 + 4x$ and the x -axis.

Classified Practice

Applications of Integration to Geometry

Q011 Written

Solve the given cubic area exercises by finding every x -intercept and adding the absolute signed pieces.

Q012 Written

Evaluate $\int_{-2}^1 |x + 1| dx$.

Classified Practice

Applications of Integration to Geometry

Q013 Written

Evaluate $\int_0^3 |x^2 - 4| dx$.

Q014 Written**Evaluate** $\int_0^2 |x^2 + 3x - 4| dx$.

Classified Practice

Applications of Integration to Geometry

Q015 Written

Find the area bounded by $y = x^2 - x$ and its tangents at $(0, 0)$ and $(2, 2)$.

Q016 Written**Find the area bounded by $y = x^3 + x^2 - 3x + 6$ and its tangent at $(1, 5)$.**

Classified Practice

Applications of Integration to Geometry

Q017 **Written**

For $y = x^2$, tangents through $(3, 8)$ touch at $x = 2, 4$. Find the enclosed area.

Q018 Written

If $y = kx$ bisects the area under $y = 2x - x^2$ and above the x -axis, find k .

Classified Practice

Applications of Integration to Geometry

Q019 **Written**

Solve the **given** bisection exercises for the given parabolas and lines, using the required area ratio.

Q020 MCQ**When the graph is below the x-axis, area is:****Choose the correct option.**

- A the signed integral
- B the negative of the integral
- C always zero
- D undefined

Classified Practice

Applications of Integration to Geometry

Q021 MCQ

A bounded area under a curve is found by first checking:**Choose the correct option.**

- A the graph and sign
- B only the y-intercept
- C the font size
- D the derivative only

Q022 MCQ

The area under $y = -x^2 + 4$ between its roots is:**Choose the correct option.**A $8/3$ B $16/3$ C $32/3$ D $64/3$

Classified Practice

Applications of Integration to Geometry

Q023 MCQ

Between two curves, the integrand is:**Choose the correct option.**

- A lower-upper
- B upper-lower
- C product
- D derivative

Q024 MCQ**If curves cross inside the interval, area must be:****Choose the correct option.**

- A split at the crossing
- B ignored
- C doubled always
- D negative

Classified Practice

Applications of Integration to Geometry

Q025 MCQ

The area between $y = x^2 - 2$ and $y = -2x + 1$ is:

Choose the correct option.

A $16/3$

B $32/3$

C $64/3$

D $9/2$

Q026 MCQ**For cubic area, the sign changes at:****Choose the correct option.**

- A every turning point
- B each x-intercept
- C the y-intercept only
- D the midpoint

Classified Practice

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Q027 MCQ

The area for $y = -x^3 + 4x$ is:**Choose the correct option.**

A 4

B 6

C 8

D 12

Q028 MCQ**A negative lobe contributes:****Choose the correct option.**

- A negative area
- B positive absolute area
- C no area
- D a derivative

Classified Practice

Applications of Integration to Geometry

Q029 MCQ

For an absolute-value integral, first locate:**Choose the correct option.**

- A the roots
- B the vertex only
- C the y-axis
- D the tangent

Q030 MCQ

The value of $\int_0^2 |x^2 + 3x - 4| dx$ is:

Choose the correct option.

A 3

B 4

C 5

D 6

Classified Practice

Applications of Integration to Geometry

Q031 MCQ

If f changes sign at c , split the integral:**Choose the correct option.**A at c

B at 0 only

C never

D after doubling

Q032 MCQ**A tangent line at $x=a$ has slope:****Choose the correct option.****A** $f(a)$ **B** $f'(a)$ **C** a^2 **D** the area

Classified Practice

Applications of Integration to Geometry

Q033 MCQ

For two tangent lines, split at their:**Choose the correct option.**

- A intersection x-coordinate
- B y-intercept only
- C vertex only
- D origin

Q034 MCQ

The area in the given parabola-tangent Warm Up is:**Choose the correct option.**A $\frac{1}{2}$

B 1

C $\frac{3}{2}$

D 3

Classified Practice

Applications of Integration to Geometry

Q035 MCQ

Bisection means each part has:**Choose the correct option.**

- A the same area
- B the same slope
- C the same root
- D zero area

Q036 MCQ**For $y=2x-x^2$, the total area above the x-axis is:****Choose the correct option.****A** 1**B** 2**C** 3**D** 4

Classified Practice

Applications of Integration to Geometry

Q037 MCQ

The line $y=kx$ must satisfy the stated:**Choose the correct option.**

- A area ratio
- B font ratio
- C root count
- D period

Q038 MCQ**In a bisection problem, the cut-off area is:****Choose the correct option.**

- A the total area
- B half the total area
- C twice the total area
- D zero

Classified Practice

Applications of Integration to Geometry

Quiz MCQ 1 **Concept Quiz · MCQ**

The area under $y = -x^2 + 4$ and above the x -axis is:

Choose the correct option.

A $16/3$

B 8

C $32/3$

D $64/3$

Quiz MCQ 2 Concept Quiz · MCQ

The area between $y = x^2 - 2$ and $y = -2x + 1$ is:

Choose the correct option.

A $16/3$

B $32/3$

C $64/3$

D $9/2$

Classified Practice

Applications of Integration to Geometry

Quiz MCQ 3 Concept Quiz · MCQ

Evaluate $\int_{-2}^1 |x + 1| dx$:

Choose the correct option.

- A 2
- B $5/2$
- C 3
- D $7/2$

Quiz MCQ 4 Concept Quiz · MCQ

The area enclosed by $y = x^2$ and tangents through (3, 8) is:

Choose the correct option.

A $1/3$

B $1/2$

C $2/3$

D $4/3$

Classified Practice

Applications of Integration to Geometry

Quiz WRITTEN 5 **Concept Quiz · Written**

Find the area bounded by $y = x^3 - 5x^2 + 6x$ and the x -axis.

Quiz WRITTEN 6 Concept Quiz · Written

Evaluate $\int_0^2 |x^2 + 3x - 4| dx$.

Classified Practice

Applications of Integration to Geometry

Quiz WRITTEN 7 Concept Quiz · Written

Find the area bounded by $y = x^2 - x$ and its tangents at $(0, 0)$ and $(2, 2)$.

Quiz WRITTEN 8 Concept Quiz · Written

If $y = kx$ bisects the area under $y = 2x - x^2$, find k .

Answer key

Use the question number shown on each page.

Q001 9

Q002 $\frac{33}{2}$ Q003 $\frac{32}{3}$ Q004 $\frac{32}{3}$ Q005 $\frac{32}{3}$ Q006 $\frac{64}{3}$ Q007 $\frac{9}{2}$

Q008 30

Q009 $\frac{37}{12}$

Q010 8

Q011 Use the numbered solution for each given cubic.

Q012 $\frac{5}{2}$ Q013 $\frac{23}{3}$

Q014 5

Q015 $\frac{3}{2}$ Q016 $\frac{64}{3}$ Q017 $\frac{2}{3}$ Q018 $k = 2 - \sqrt[3]{\frac{3}{4}}$

Q019 Use the numbered solution for each given ratio.

Q020 B

Q021 A

Q022 C

Q023 B

Q024 A

Q025 B

Q026 B

Q027 C

Q028 B

Q029 A

Q030 C

Q031 A

Q032 B

Q033 A

Q034 C

Q035 A

Q036 C

Q037 A

Q038 B

Quiz MCQ 1 C

Quiz MCQ 2 B

Quiz MCQ 3 B

Quiz MCQ 4 C

Quiz WRITTEN 5 $\frac{37}{12}$

Quiz WRITTEN 6 5

Quiz WRITTEN 7 $\frac{3}{2}$ Quiz WRITTEN 8 $2 - \sqrt[3]{\frac{3}{4}}$